

PREMIER UNIVERSITY, CHATTOGRAM

Department of Computer Science & Engineering



Final Year Design Project Proposal

On

A Consensus-Based Family Document Graph Approach for Civil Registry Inconsistency Detection and Correction in Bangladesh

Course Code: CSE 4000

Credits: 4

Semester: 4th Year

Session: Spring 2026

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May 1, 2026

What This Project Is About

In Bangladesh, civil documents like the National Identity Card (NID), Birth Certificate, and Academic Certificate are very important for every person. A very common problem is that the same person's name, date of birth, or address is written in different ways in different documents. This causes problems at government offices, job applications, and passport work.

The problem gets worse in a family. A child's documents must show the same parent names that are in the parents' own documents. If the child's Father Name field does not match the father's NID name, the family must go through a long correction process.

Right now, the government correction process works by hand. A user applies to fix one document at a time and an officer checks it manually. There is no system that looks at all family documents together and finds all problems before the correction is applied.

This project builds a system that:

1. Takes all family member document data from an Excel file.
2. Finds every place where each person's name is written across all family documents.
3. Counts which name version appears most and uses that as the correct value.
4. Finds small name mistakes and serious name mismatches.
5. Checks dates, parent-child age gaps, and address spellings.
6. Gives a mismatch report and fix suggestions in an Excel output file.

The Main New Idea

The main new idea is the **Family Document Graph with a Consensus Method**.

In a family, one person's name appears in many places. For example, the father's name is written in:

Document / Field	Name Written
Father's own NID	Md. Ali
Father's own Birth Certificate	Mohammad Ali
Father's own Academic Certificate	Md. Ali
Son's Birth Certificate — Father Name	Mohammad Ali
Son's NID — Father Name	Md. Ali
Son's Certificate — Father Name	Md. Ali
Daughter's Birth Certificate — Father Name	Mohammad Ali
Daughter's NID — Father Name	Md. Ali

The system collects all these values and counts them:

- **“Md. Ali”** — 5 times → Most common → **This is the correct value**
- **“Mohammad Ali”** — 3 times → Small name mistake

Fix suggestion: All documents should use “Md. Ali”.

This same method is used for the mother’s name and each child’s own name across all their documents. No current system does this.

What the System Checks

Name Check

For every person in the family, the system finds all written versions of their name from all related family documents, counts each version, and picks the most common one. Other versions are marked as:

- **Small Mistake** — Only a short form or spelling difference. Example: “Md. Ali” vs. “Mohammad Ali”.
- **Serious Mismatch** — The main name is completely different. Example: “Mohammad Ali” vs. “Md. Akib”. This needs a senior officer to check.

Date of Birth Check

All date values are first converted to one standard format. Then the real dates are compared. Two cases are possible:

- **Format difference only:** “01/05/1995” and “1995-05-01” are the same real date. This is **not a mismatch**.
- **Real date difference:** “2002-03-03” in NID and “2000-02-02” in Birth Certificate are two different dates. This is a **true DOB mismatch** and needs officer verification.

Parent-Child Age Gap Check

A parent must be older than their child by a proper number of years. The system checks the real year difference between parent and child DOB. If the gap is too small or invalid, the system marks it as a problem.

- **Valid example:** Father born 1980, Son born 2005 — gap is 25 years. **Valid**.
- **Invalid example:** Father born 1980, Child born 1982 — gap is only 2 years. **Invalid**.

Address Check

Division, District, Upazila, Union, City Corporation, and Pourosova names are checked against a fixed spelling list. If a document uses a wrong spelling, the system suggests the correct one. For example, if a document writes “Chittagong” but the fixed spelling is “Chattogram”, the system suggests “Chattogram”.

Why This Is Research

Research Question

Does a consensus-based Family Document Graph give better results than the current manual single-correction process for finding and fixing name problems in Bangladesh civil registry documents?

Research Gap

No current system does what this project does:

1. No system collects one person's name from all related family documents and uses a counting method to find the correct value.
2. The current government correction process works one application at a time. It does not find all family-level problems at once.
3. No system separates small name mistakes from serious mismatches automatically.
4. No system checks the parent-child age gap across all related documents.
5. No Bangladesh-specific automatic system exists for this problem.

How the Research Will Be Done

1. **Make a test dataset:** 50 to 100 family records in Excel files, with different types of name mistakes, date problems, age gap problems, and address problems. The correct answer for each record will be marked by hand.
2. **Build the old method:** The current one-at-a-time correction process will be built and tested on the same dataset as a baseline.
3. **Build the new method:** The Family Document Graph and Consensus Method will be built and tested on the same dataset.
4. **Compare both:** The results will be compared using Accuracy, Precision, Recall, and F1-Score to show which method works better.

Known Limitations

Every system has cases where it may not work well. The following cases are known limitations of the proposed method.

When Most Documents Have the Wrong Value

The consensus method picks the name that appears most often. If most documents in the family contain the wrong name, the system will pick that wrong name as the correct value.

Example	What Happens
Father's NID: Md. Ali Son's Birth F.Name: Karim Hos- sain Son's NID F.Name: Karim Hos- sain Daughter's NID F.Name: Karim Hossain	"Karim Hossain" wins the count (3 vs 1). System picks a wrong value as correct.

Partial fix: The system gives extra weight to the Father's own NID field, since this is the most trusted document.

When Only One Document Exists

If a person has only one document, the system cannot count and compare. There is nothing to check against. The system will mark this as "not enough data" instead of giving a suggestion.

When Bengali and English Are Mixed

The system may count the Bengali script version and "Mohammad Ali" (English) as two different names, even though they are the same name. This can lower the count for the correct value.

Partial fix: Bengali name values will be converted to English using a simple transliteration step before counting.

When the Child's Parent Name Is Completely Different

If a child's document shows a completely different name in the Father Name field (not just a spelling difference), the system marks this as a serious mismatch and asks for officer verification. However, the system cannot tell *why* the name is different — it could be a data entry error or a genuinely different person.

Summary of known failure cases:

Case	System Result	Partial Fix
Wrong name in most documents	Wrong consensus	Weight NID more
Only one document per person	No suggestion	Mark as low data
Bengali and English mixed	Low count	Transliterate first
Child doc shows fully different parent name	Serious mismatch flag	Officer check
Two different real DOB values	True DOB mismatch	Officer check

These limitations will be discussed in the final report and can become areas for future improvement.

Input and Output

Input

Family data will be entered through an **Excel file**. Each row contains one person's document data. The Excel file will have these columns:

Family ID, Person ID, Role (Father / Mother / Son / Daughter), NID Name, Birth Certificate Name, Academic Certificate Name, Father Name, Mother Name, Date of Birth, Division, District, Upazila, Union, City Corporation or Pourosova, and Address details.

Output

The system will give two outputs:

- **Excel report file** — Shows the correct value, mismatch type, and fix suggestion for each field and each person.
- **PDF summary report** — A clear summary of all found problems and suggestions for the whole family.

Sample Output Table

Person / Field	All Values Found	Correct Value	Result
Father — Name	NID: Md. Ali Birth: Mohammad Ali Son Birth F: Mohammad Ali Son NID F: Md. Ali Daughter NID F: Md. Ali	Md. Ali (5/8)	Small Mistake
Father — DOB	NID: 2002-03-03 Birth: 2000-02-02	Cannot decide	True DOB Mismatch
Age Gap	Father: 1980 Child: 1982	2 years	Invalid
Address — District	NID: Chittagong Birth: Chattogram	Chattogram	Spelling Fix
Son — Father Name	Son Birth F: Karim Hossain Father NID: Md. Ali	Cannot decide	Serious Mismatch

Tools

Data Input: Excel file (.xlsx)

Processing: Python — counting engine, name checker, date fixer, age gap checker, address checker

Database: MySQL / PostgreSQL

Output: Excel report file + PDF summary (ReportLab)

Testing: Python — Accuracy, Precision, Recall, F1-Score

Project Timeline

The project runs over two semesters.

Semester 1 (CSE 4000A — 1 Credit) — Planning

Phase	Task	Time
Phase 1	Topic Study and Reading	Week 1–3
Phase 2	Database and System Design	Week 4–6
Phase 3	Collect and Prepare Excel Test Dataset	Week 7–9
Phase 4	Proposal Writing and Submission	Week 10–12

Semester 2 (CSE 4000B — 3 Credits) — Main Work

Phase	Task	Time
Phase 5	Build Old Method (One-at-a-time correction)	Week 1–2
Phase 6	Build Family Document Graph and Counting Engine	Week 3–6
Phase 7	Build Name, Date, Age Gap, and Address Checkers	Week 7–9
Phase 8	Build Excel and PDF Report Output	Week 10–11
Phase 9	Test Both Methods and Compare Results	Week 12–13
Phase 10	Final Report Writing and Defense Practice	Week 14–16

Conclusion

This project builds a new system that finds name problems, date problems, age gap problems, and address problems in Bangladesh civil documents at the family level. The main new idea is to collect every version of a person's name from all family documents, count which version appears most, and use that as the correct value. The project also knows its own limits — for example, when most documents contain the wrong value, or when only one document exists, the system may not give a good result. These limits will be discussed in the final report. This project covers Database Design, Software Engineering, and Data Science. It works on a real problem that affects millions of people in Bangladesh, making it a good fit for the Final Year Design Project of the B.Sc. in CSE program at Premier University Chittagong.

Signature of Group Leader

Name: _____

Date: _____

Signature of Supervisor

Name: _____

Date: _____